SOURCE TO SOUND

CLEAN THE WATER CHALLENGE

Compete to create the best water filter!

WHAT IS STORMWATER?

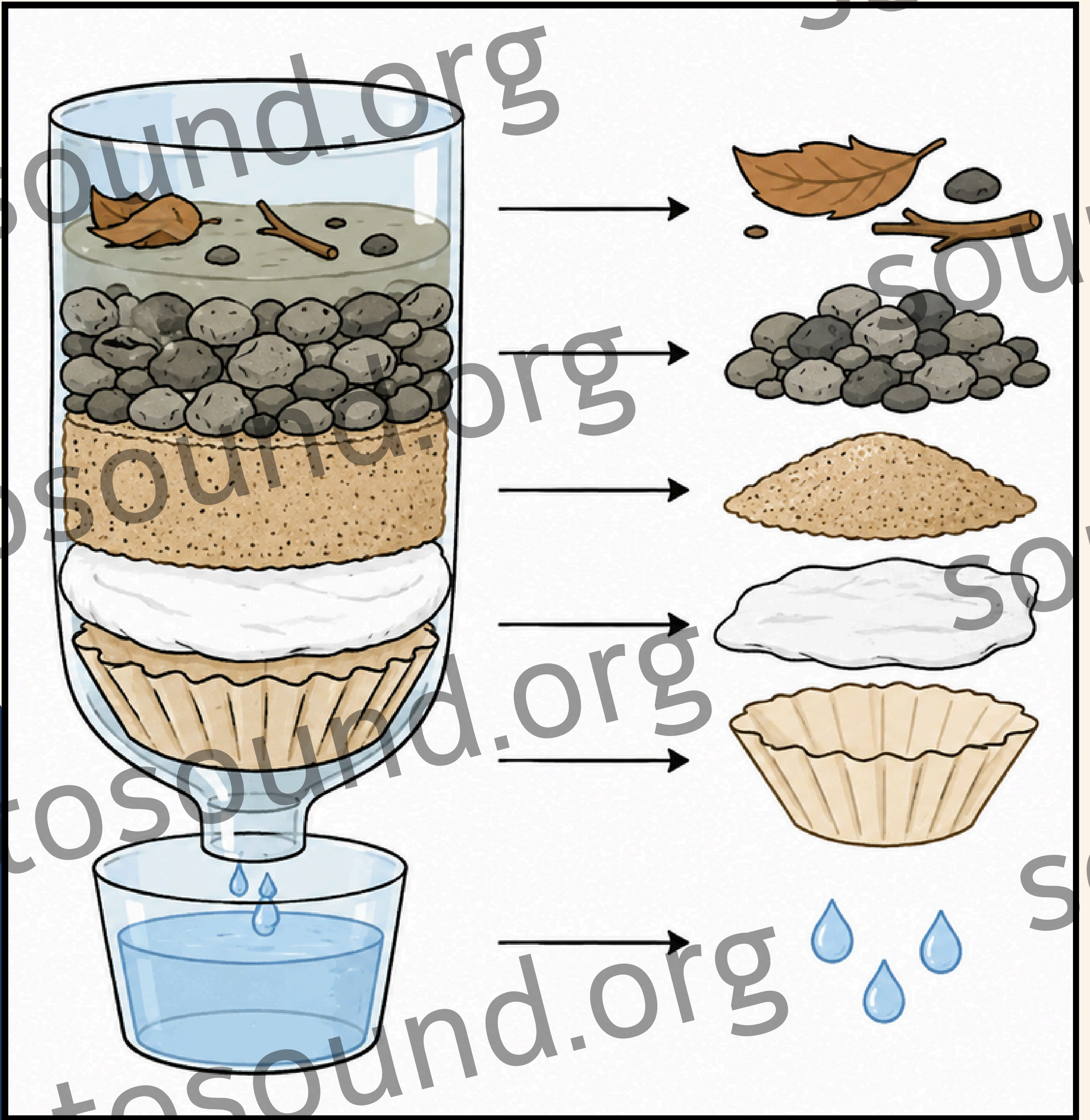
When it rains, water flows across roads and surfaces picking up dirt, oil, and trash along the way. This dirty water is called stormwater.

It flows straight into the nearest stream or river without being cleaned. That stream might be home to salmon, frogs, and insects that need clean water to survive.

WHAT IS FILTRATION?

Filtration is the process of removing particles and pollution from water by passing it through different materials. Each material traps a different type of particle. Layers of different materials working together can clean water more effectively than any single material alone.

HOW DO DIFFERENT MATERIALS FILTER WATER?



Gravel catches the biggest particles like rocks, leaves, and large debris.

Sand catches smaller particles that slip through the gravel.

Cotton and coffee filters catch the tiniest particles that sand misses.

YOUR CHALLENGE

Today your team is going to build a water filter using the same materials that real water treatment plants use. You will pour dirty water through your filter and compete to see which team can produce the clearest, cleanest water in the fastest time!

TIP

Think carefully about which materials you choose and the order you stack them!

MATERIALS

- 1 plastic water bottle cut in half
- Pebbles
- Gravel
- Sand
- Cotton balls
- Coffee filter

You have 15 minutes to fill the top half of your bottle with your chosen filter materials in whatever order you think will work best. When time is up, we will pour the same amount of dirty water through every team's filter at the same time!

TESTING TIME!

We are judging on clarity, speed, and
final volume of water.

DISCUSSION

In your groups, discuss:

Which material do you think did the most work in your filter? Why?

What would happen if you only used one material instead of layers?

What other materials from real life do you think could work as a filter?

If you could redesign your filter knowing what you know now, what would you change?

Where else in your neighborhood do you think water gets filtered before it reaches a stream?